UNIVERSITY OF THE PEOPLE

MATH 1280-001-AY2026-T4

Assignment Activity Unit 7

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Computing Probabilities and Percentiles Using the Normal Distribution

Introduction

Statistical tools such as the normal distribution allow researchers and social workers to interpret large datasets and draw meaningful conclusions about communities. In the scenario presented, household income data collected from families in a rural, underprivileged community follows a normal distribution with a mean (μ) of \$3,000 and a standard deviation (σ) of \$500. Using Z-scores and the standard normal distribution table, it is possible to calculate the probability that a randomly selected family earns below a specific threshold and to determine the percentile rank of families with higher incomes. These calculations provide critical insights for effective resource allocation and policy planning.

Question 1: Probability That a Family Has an Income Below \$2,500

To find the probability that a randomly selected family has an income below $2,500, the Z-score must first be calculated. The Z-score formula converts a raw data value into a standardized value that indicates how many standard deviations the value lies from the mean (Illowsky et al., 2023). The formula is as follows:

$$\mathbf{Z = (x - \mu) / \sigma}$$

Where:

x = observed value = $2,500

μ = mean = $3,000

σ = standard deviation = $500

Step 1: Substituting the values into the formula:

$$\mathbf{Z = (2{,}500 - 3{,}000) / 500}$$

$$\mathbf{Z = -1}$$

Step 2: Finding the probability using the standard normal distribution table:

A Z-score of −1 indicates that an income of $2,500 is one standard deviation below the mean. Using the standard normal distribution table (University of Arizona, n.d.), the cumulative probability to the left of Z = −1 is:

$$\mathbf{P(X < 2{,}500) = P(Z < -1) = 0.1587}$$

This result means that approximately 15.87% of families in the community earn less than $2,500 per month. From a social work perspective, this proportion is significant because it identifies the share of households that may be most vulnerable and in need of targeted financial assistance or welfare services. As Illowsky et al. (2023) explain, probabilities derived from the normal distribution allow professionals to identify and address the needs of specific population segments more effectively.

Question 2: Z-Score and Percentile Rank for an Income of $3,500

For a family with an income of $3,500, the same Z-score formula is applied to determine how this income compares to the rest of the distribution. Percentile rank expresses the percentage of

observations that fall below a given value, making it a powerful tool for comparing individual data points within a dataset (Illowsky et al., 2023).

Step 1: Calculate the Z-score for x = $3,500:

$$Z = (3{,}500 - 3{,}000) / 500$$

$$Z = 500 / 500 = 1$$

Step 2: Find the percentile rank using the Z-table:

A Z-score of 1 indicates that an income of $3,500 is one standard deviation above the mean. Using the standard normal distribution table (University of Arizona, n.d.), the cumulative probability to the left of Z = 1 is:

$$P(Z < 1) = 0.8413 \rightarrow 84.13\text{th Percentile}$$

This result means that a family earning $3,500 per month earns more than approximately 84.13% of all families in the community. In terms of resource allocation, this family falls in the upper income tier of this rural community and would generally not be prioritized for financial aid compared to households in the lower percentiles. Understanding percentile ranks within income distributions helps social workers and policymakers focus limited resources on the most economically disadvantaged households.

Practical Implications for Resource Allocation

The normal distribution is a powerful tool in applied statistics precisely because it allows analysts to make data-driven decisions about populations. According to Illowsky et al. (2023), the empirical rule, also known as the 68, 95, 99.7 rule, states that approximately 68% of values in a normal distribution fall within one standard deviation of the mean, about 95% within two standard deviations, and approximately 99.7% within three standard deviations. In this income scenario, that means:

- About 68% of families earn between $2,500 and $3,500 (within ± 1 SD of the mean).
- About 95% earn between $2,000 and $4,000 (within ± 2 SD).
- Families earning below $2,000 or above $4,000 represent extreme cases in either tail.

By combining Z-scores, probabilities, and percentile ranks, social workers and community planners can objectively identify which households fall below income thresholds and estimate the proportion of the population requiring immediate intervention. This approach transforms raw data into actionable insights that support equitable, evidence-based decision-making in underserved communities.

Conclusion

This assignment demonstrated how the normal distribution can be used to compute meaningful probabilities and percentile ranks from income data. Using a mean of $3,000 and a standard deviation of $500, it was determined that approximately 15.87% of families earn below $2,500 ($Z = -1$), and that a family earning $3,500 falls at the 84.13th percentile ($Z = 1$). These calculations highlight the practical value of statistical reasoning in social work and community planning, where understanding the distribution of income is essential for targeting resources appropriately and improving outcomes for the most vulnerable populations.

References

Illowsky, B., Dean, S., Birmajer, D., Blount, B., Boyd, S., Einsohn, M., Helmreich, J., Kenyon, L., Lee, S., & Taub, J. (2023). Introductory statistics 2e. OpenStax. https://openstax.org/details/books/introductory-statistics-2e/

University of Arizona. (n.d.). Standard normal distribution: Table values represent the area to the left of the Z score. https://math.arizona.edu/~rsims/ma464/standardnormaltable.pdf